Harbor Tide Sensor: a synthetic field note

Abstract

This synthetic note was written for the M7a live smoke test and describes an invented sensor.

Introduction

The Harbor tide sensor is an imaginary instrument placed on a fictional pier for this test.
It records water level every five minutes and stores readings on a small invented logger.
No real harbor, person, dataset or publication is described anywhere in this document.
The note exists only so that a PDF with enough extractable body text can be indexed.
Over thirty synthetic days the Harbor tide sensor reported a mean error of 1.2 centimeters against a fictional reference gauge installed beside it on the same imaginary pier.
The error grew during storm periods and returned to its usual level within two days.
Each reading was compared with the reference gauge at the same five minute interval.
The invented logger kept every reading and flagged values outside a fixed tolerance band.
The mean error of 1.2 centimeters stayed below the tolerance band on twenty eight days.
On two storm days the error exceeded the band and those readings were flagged for review.
The sensor did not lose power during the invented deployment and no reading was missing.
Calibration was checked once at the start and once at the end of the thirty day period.
These synthetic numbers are not measurements and must not be cited as real findings.
They are only meant to give a retrieval system a second document with distinct content.
A comparison question can contrast this sensor error with the drift of another device.
The Harbor tide sensor and the Lumen calibration bench are both fictional instruments.
Further synthetic paragraphs follow to keep enough body text on this single page.
The fictional pier was described as calm on most days with light wind from the west.
Maintenance visits were imagined as weekly and lasted about twenty minutes each time.
During each visit the invented technician wiped the sensor housing and checked cables.
No firmware change was made during the deployment described in this synthetic note.
The logger clock was assumed to be correct and no time correction was applied.
All values in this note are invented for software testing and carry no real meaning.
The invented deployment log lists each storm day with its start and end time in hours.
Water temperature was not measured because the fictional sensor has no thermometer.
A second imaginary pier was considered for the test but it was never used in this note.
The tolerance band was set at two centimeters before the synthetic deployment began.
Readings flagged during storms were kept in the log rather than deleted from the record.
The synthetic summary above is the only result reported for the Harbor tide sensor.
Nothing in this document should be read as advice about real tides or real instruments.

References

[1] Synthetic Author. An invented reference used only for parser testing. 2026.